

Scalability & Future Plan

Date	05 July 2026
Team ID	XXXXXX
Project Name	Emotion Support Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	API Rate Limits: The free tier of the Gemini API limits the number of requests per minute.	If multiple users access the app simultaneously, some may experience response delays or timeout errors.	High
2	Volatile Memory: Streamlit's session_state is temporary and clears upon a page refresh.	Users will lose their current chat history if their browser reloads or connection drops.	Medium
3	Text-Only Input: The application currently relies entirely on typing.	Limits accessibility for users who are too exhausted, anxious, or physically unable to type out their feelings.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Hosted on Streamlit Community Cloud (shared resources).	Migrate hosting to a scalable cloud provider (e.g., AWS EC2 or Google Cloud Run) with auto-scaling to handle traffic spikes.
2	Data Storage	Temporary in-memory storage (session_state).	Integrate a secure NoSQL database (like Firebase or MongoDB) to save persistent, encrypted chat logs for returning users.
3	Performance	Synchronous API calls block the UI while generating.	Implement asynchronous processing (AsyncIO) to handle multiple user requests concurrently without freezing the interface.
4	Security	Basic anonymous access; API keys hidden via .toml secrets.	Implement optional OAuth2 user authentication and end-to-end encryption for stored user chat histories.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Voice-to-Text & Crisis Routing: Allowing users to speak their feelings and automatically routing severe distress keywords to human helplines.	1 - 3 Months	Massively increases accessibility for exhausted users and ensures safety during mental health emergencies.
Phase 3	Personalized Support Dashboard: Adding secure user accounts (OAuth) to save encrypted chat histories and track mood trends over time.	3 - 6 Months	Transitions the app from a one-time venting tool into a long-term mental wellness companion.
Phase 4	Native Mobile App Deployment: Converting the web application into a downloadable iOS and Android app using React Native or Flutter.	6 - 12 Months	Expands the user base globally and allows for native push notifications (e.g., daily check-ins).